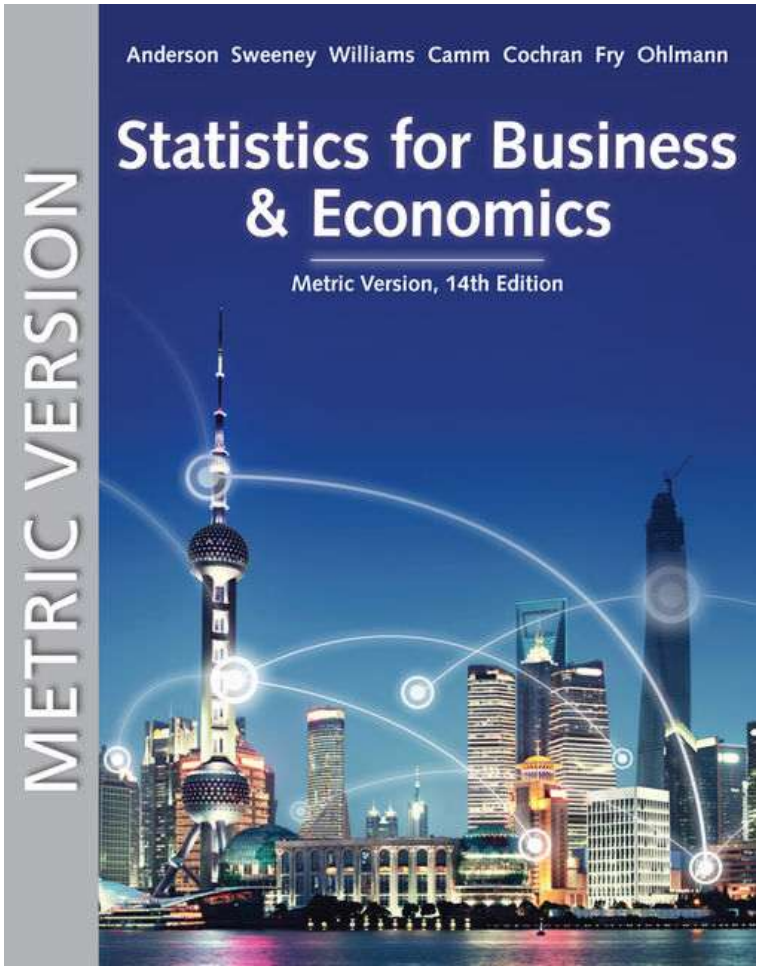
Statistics for Business and Economics (14e) Metric Version

Anderson, Sweeney, Williams, Camm, Cochran, Fry, Ohlmann

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Chapter 3 - Descriptive Statistics: Numerical Measures

3.1 - Measures of Location

3.2 - Measures of Variability

3.3 - Measures of Distribution Shape, Relative Location, and Detecting Outliers

3.4 - Five-Number Summaries and Box Plots

3.5 - Measures of Association Between Two Variables

3.6 - Data Dashboards: Adding Numerical Measures to Improve Effectiveness

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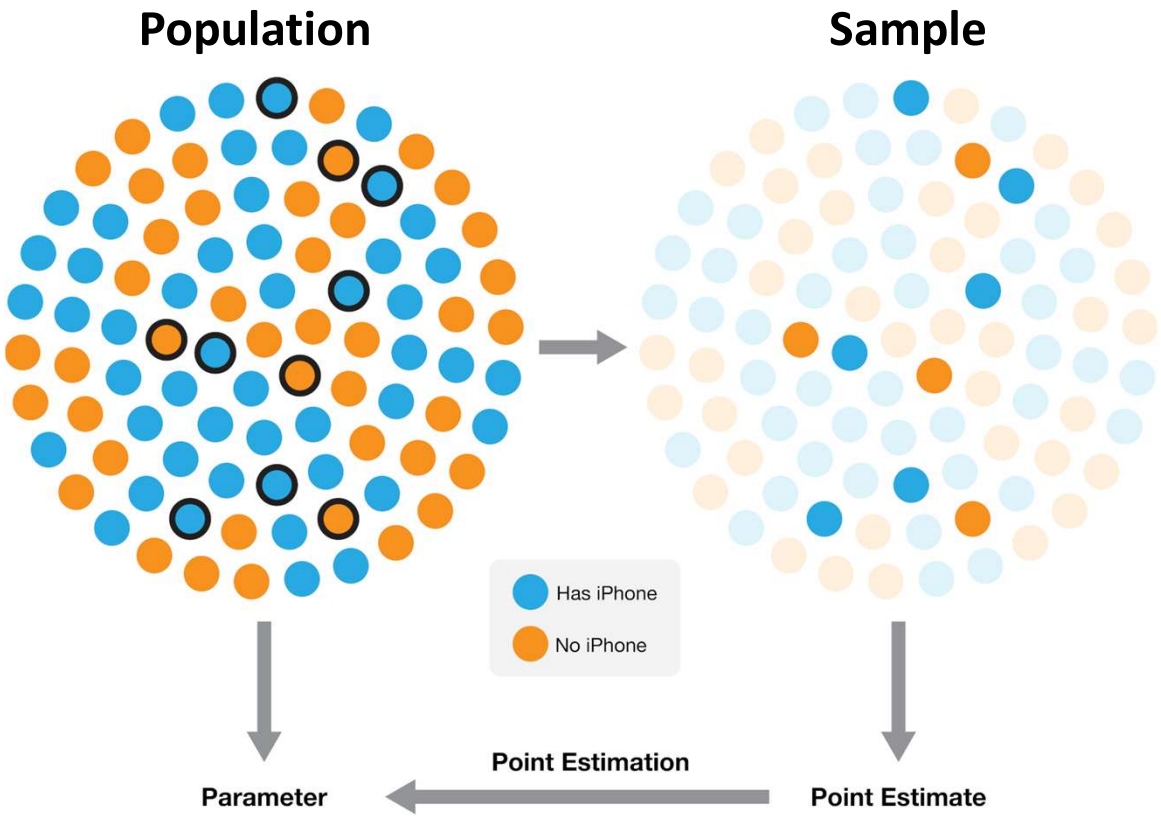
Numerical Measures

- If the measures are computed for data from a sample, they are called sample statistics.
- If the measures are computed for data from a population, they are called population parameters.
- A sample statistic is referred to as the point estimator of the corresponding population parameter.



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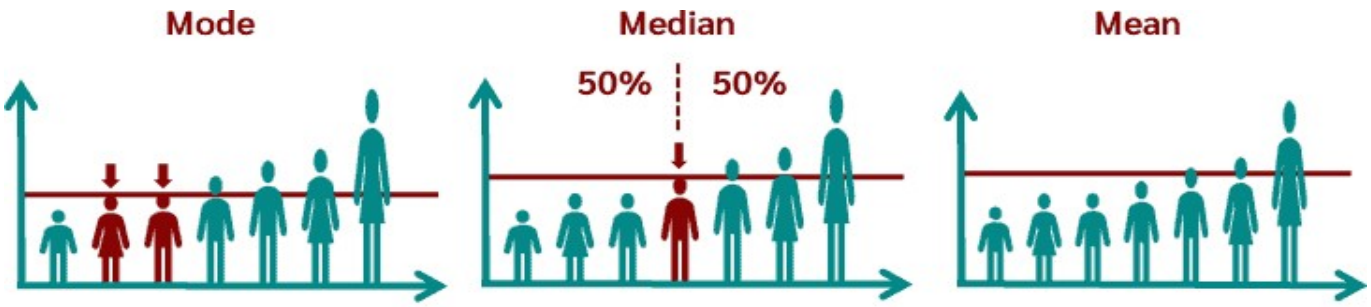
Statistical Inference



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Measures of Location

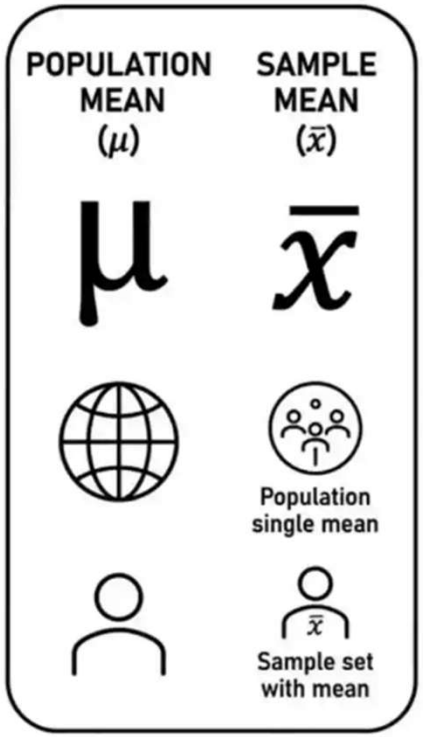
- Mean
- Median
- Mode
- Weighted Mean
- Geometric Mean
- Percentiles
- Quartiles



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Mean

- The mean provides a measure of central location.
- The mean of a data set is the average of all the data values.
- The sample mean $\bar{x}$ is the point estimator of the population mean, μ.



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Sample Mean $\bar{x}$

$$\bar{x} = \frac{\sum x_i}{n}$$

where $\sum x_i$ = the sum of the values of the *n* observations and

n = the number of observations in the sample.

"Σ" means "add them all together"

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Sample Mean $\bar{x}$

Seventy efficiency apartments were randomly sampled in a college town. The monthly rents for these apartments are listed below.

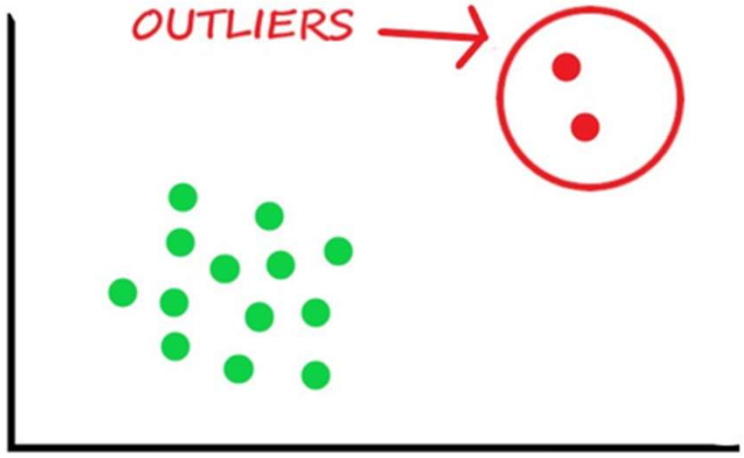
545	715	530	690	535	700	560	700	540	715
540	540	540	625	525	545	675	545	550	550
565	550	625	550	550	560	535	560	565	580
550	570	590	572	575	575	600	580	670	565
700	585	680	570	590	600	649	600	600	580
670	615	550	545	625	635	575	650	580	610
610	675	590	535	700	535	545	535	530	540

$$\bar{x} = \frac{\sum x_i}{n} = \frac{41{,}356}{70} = 590.80$$

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Median

- The median of a data set is the value in the middle when the data items are arranged in ascending order.
- Whenever a data set has extreme values (i.e., outliers), the median is the preferred measure of central location.



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Median

- The median is the measure of location most often reported for annual income and property value data.
- A few extremely large incomes or property values can inflate the mean.



Example:
The annual incomes of 10 people, measured in thousands of dollars: 35, 40, 45, 50, 55, 60, 65, 70, 75, and 500

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Median

Here we have an odd number of observations:

7 observations: 26, 18, 27, 12, 14, 27, and 19.

Rewritten in ascending order: 12, 14, 18, 19, 26, 27, and 27.

The median is the middle value in this list, so the median = 19.

Median Position = (n + 1) ÷ 2

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Median

Here we have an even number of observations:

8 observations: 26, 18, 27, 12, 14, 27, 19, and 30.

Rewritten in ascending order: 12, 14, 18, 19, 26, 27, 27, and 30.

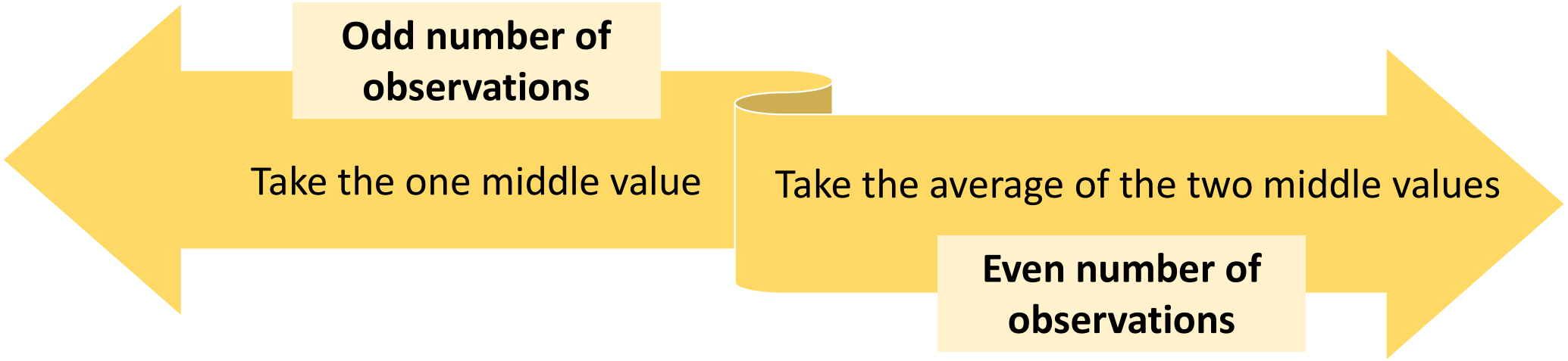
The median is the average of the two middle values in this list

➔ So the median = (19 + 26)/2 = 22.5.

- The median does not have to be one of the actual values in the data set.

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Median



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Median

Example: Apartment Rents

Mean➔ $\bar{x} = \frac{\sum x_i}{n} = \frac{41{,}356}{70} = 590.80$

Median➔ ?

545	715	530	690	535	700	560	700	540	715
540	540	540	625	525	545	675	545	550	550
565	550	625	550	550	560	535	560	565	580
550	570	590	572	575	575	600	580	670	565
700	585	680	570	590	600	649	600	600	580
670	615	550	545	625	635	575	650	580	610
610	675	590	535	700	535	545	535	530	540

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Median

Example: Apartment Rents

Notice that there are 70 values provided which are in ascending order.

Averaging the 35th and 36th values: Median ➔(575 + 575)/2 = 575.

525	530	530	535	535	535	535	535	540	540
540	540	540	545	545	545	545	545	550	550
550	550	550	550	550	560	560	560	565	565
565	570	570	572	575	575	575	580	580	580
580	585	590	590	590	600	600	600	600	610
610	615	625	625	625	635	649	650	670	670
675	675	680	690	700	700	700	700	715	715

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Mode (眾數)

- The mode of a data set is the value that occurs with greatest frequency.
- The greatest frequency can occur at two or more different values.
  - If the data have exactly two modes, the data are bimodal (雙峰).
  - If the data have more than two modes, the data are multimodal (多峰).

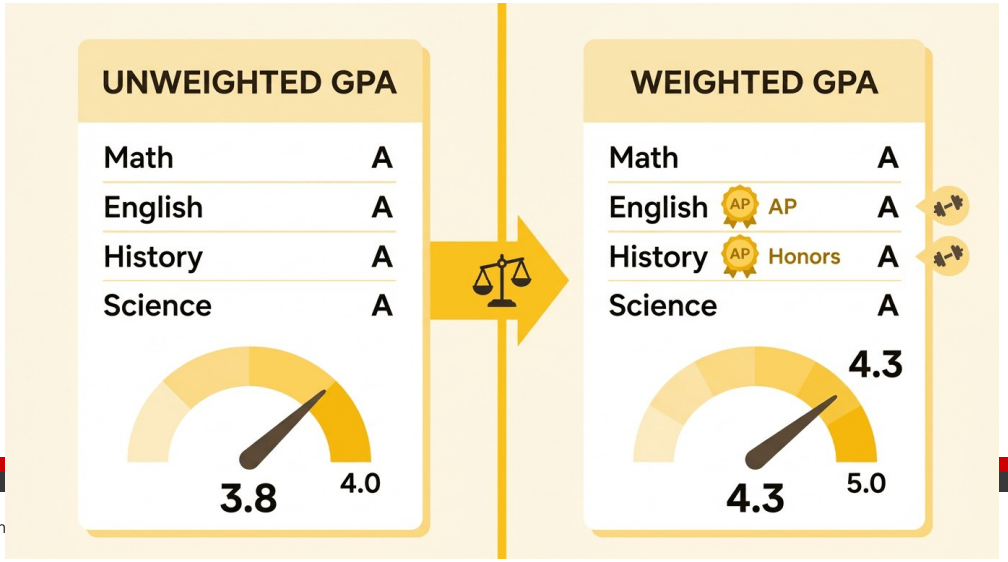
The mode is 550.

525	530	530	535	535	535	535	535	540	540
540	540	540	545	545	545	545	545	550	550
550	550	550	550	550	560	560	560	565	565
565	570	570	572	575	575	575	580	580	580
580	585	590	590	590	600	600	600	600	610
610	615	625	625	625	635	649	650	670	670
675	675	680	690	700	700	700	700	715	715

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Weighted Mean (加權平均數)

- In some instances, the mean is computed by giving each observation a weight that reflects its relative importance.
- The choice of weights depends on the application.
- The weights might be the number of credit hours earned for each grade, as in GPA.
- In other weighted mean computations, quantities such as kilograms, dollars, or volume are frequently used.



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Weighted Mean (加權平均數)

$$\bar{x} = \frac{\sum w_i x_i}{\sum w_i}$$

where: x_i = value of observation i

w_i = weight for observation i

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Weighted Mean

Andy Butler, a home builder, is looking over the expenses he incurred for a house he just built. For the purpose of pricing future projects, he would like to know the average wage ($/hour) he paid the workers he employed. Listed below are the categories of workers he employed, along with their respective wage and total hours worked.

Worker	Wage ($/hr)	Total Hours
Carpenter	21.60	520
Electrician	28.72	230
Laborer	11.80	410
Painter	19.75	270
Plumber	24.16	160

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Weighted Mean

Example: Construction Wages

Worker	x_i	w_i	$w_i x_i$
Carpenter	21.60	520	11232.0
Electrician	28.72	230	6605.6
Laborer	11.80	410	4838.0
Painter	19.75	270	5332.5
Plumber	24.16	160	3865.6
		1590	31873.7

$$\bar{x} = \frac{\sum w_i x_i}{\sum w_i} = \frac{31{,}873.7}{1{,}590} = 20.0464 = \$20.05$$

FYI, the equally-weighted (simple) mean = $21.21

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Geometric Mean (幾何平均數)

- The geometric mean is calculated by finding the *n*th root of the product of *n* values.
- It is often used in analyzing growth rates in financial data (where using the arithmetic mean will provide misleading results).
- It should be applied anytime you want to determine the mean rate of change over several successive periods (be it years, quarters, weeks, . . .).
- Other common applications include: changes in populations of species, crop yields, pollution levels, and birth and death rates.

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Geometric Mean

$$\bar{x}_g = \sqrt[n]{(x_1)(x_2)\dots(x_n)}$$

$$= [(x_1)(x_2)\dots(x_n)]^{1/n}$$



Note

- Arithmetic Mean ➔Additive situations
- Geometric Mean ➔Multiplicative or compounding situations

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Geometric Mean

Growth Factor = 1 + Rate of Return

Example: Rate of Return

Period	Return (%)
1	-6.0
2	-8.0
3	2.0
4	5.4



Growth Factor
0.940
0.920
1.020
1.054

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Geometric Mean

Example: Rate of Return

Period	Return (%)
1	-6.0
2	-8.0
3	2.0
4	5.4



Growth Factor
0.940
0.920
1.020
1.054

$$\bar{x}_g = \sqrt[5]{(0.94)(0.92)(1.02)(1.054)} = 0.9819$$

The average growth rate per period is (0.9819 – 1)(100) = –1.81%.

Average Growth Factor = (Geometric mean − 1) x 100%

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Percentiles

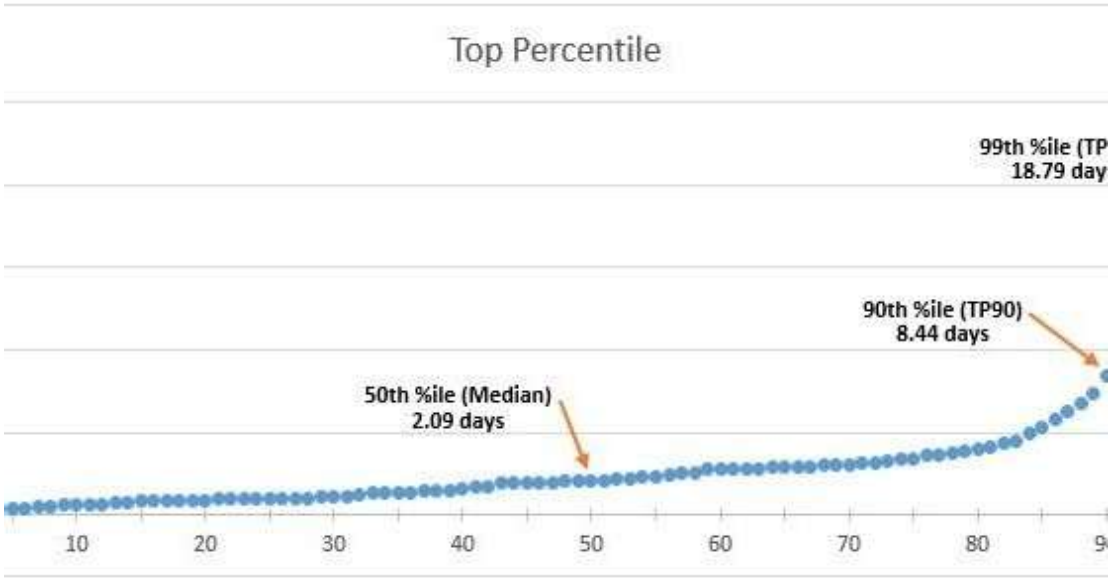
- A percentile provides information about how the data are spread over the interval from the smallest value to the largest value.
- Percentiles tell us the relative position of a value within an ordered data set
- Admission test scores for colleges and universities are frequently reported in terms of percentiles.
- The pth percentile of a data set is a value such that at least p percent of the items take on this value or less and at least $(100 - p)$ percent of the items take on this value or more.
  - 50th percentile ➔ About 50% of the observations are at or below it, and about 50% are at or above it

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Percentiles

- Arrange the data in ascending order.
- Compute L_p, the location of the pth percentile.

$$L_p = \left(\frac{p}{100}\right)(n+1)$$



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80th Percentile

Example: Apartment Rents $L_p = \left(\frac{p}{100}\right)(n+1) = \left(\frac{80}{100}\right)(70+1) = 56.8$

525	530	530	535	535	535	535	535	540	540
540	540	540	545	545	545	545	545	550	550
550	550	550	550	550	560	560	560	565	565
565	570	570	572	575	575	575	580	580	580
580	585	590	590	590	600	600	600	600	610
610	615	625	625	625	635	649	650	670	670
675	675	680	690	700	700	700	700	715	715

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80th Percentile

Example: Apartment Rents $L_p = \left(\frac{p}{100}\right)(n+1) = \left(\frac{80}{100}\right)(70+1) = 56.8$

The 80th percentile is the 56th value plus 0.8 times the difference between the 57th and 56th values.

So the 80th percentile = 635 + 0.8(649 – 635) = 646.2.

525	530	530	535	535	535	535	535	540	540
540	540	540	545	545	545	545	545	550	550
550	550	550	550	550	560	560	560	565	565
565	570	570	572	575	575	575	580	580	580
580	585	590	590	590	600	600	600	600	610
610	615	625	625	625	635	649	650	670	670
675	675	680	690	700	700	700	700	715	715

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80th Percentile, Part 2

Example: Apartment Rents

"At least 80% of the items take on a value of 646.2 or less."

"At least 20% of the items take on a value of 646.2 or more."

56/70 = .8 or 80%

14/70 = .2 or 20%

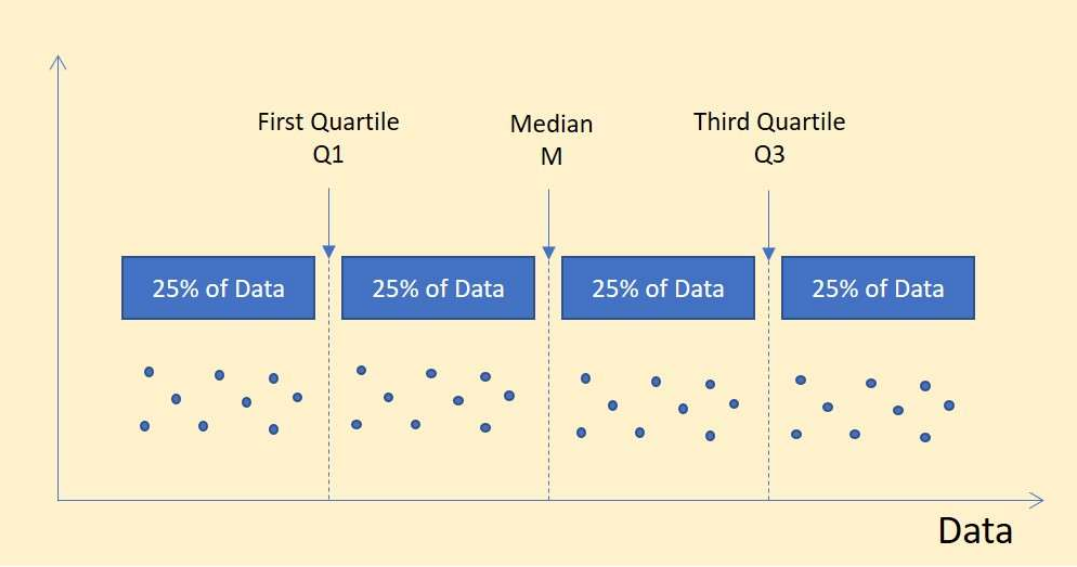
525	530	530	535	535	535	535	535	540	540
540	540	540	545	545	545	545	545	550	550
550	550	550	550	550	560	560	560	565	565
565	570	570	572	575	575	575	580	580	580
580	585	590	590	590	600	600	600	600	610
610	615	625	625	625	635	649	650	670	670
675	675	680	690	700	700	700	700	715	715

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Quartiles (四分位數)

Quartiles are specific percentiles.

1. First Quartile = 25th Percentile
2. Second Quartile = 50th Percentile = Median
3. Third Quartile = 75th Percentile



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Third Quartile (75th Percentile)

Example: Apartment Rents

$$L_p = \left(\frac{p}{100}\right)(n+1) = \left(\frac{75}{100}\right)(70+1) = 53.25$$

The 75th percentile is the 53rd value plus 0.25 times the difference between the 54th and 53rd values.

The 75th percentile = third quartile = 625 + 0.25(625 – 625) = 625.

525	530	530	535	535	535	535	535	540	540
540	540	540	545	545	545	545	545	550	550
550	550	550	550	550	560	560	560	565	565
565	570	570	572	575	575	575	580	580	580
580	585	590	590	590	600	600	600	600	610
610	615	625	625	625	635	649	650	670	670
675	675	680	690	700	700	700	700	715	715

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Measures of Variability (離散量數/變異量數)

- It is often desirable to consider measures of variability (dispersion), as well as measures of location.
- For example, in choosing supplier A or supplier B we might consider not only the average delivery time for each, but also the variability in delivery time for each.
- Common measures of variability are:
  - Range
  - Interquartile Range
  - Variance
  - Standard Deviation
  - Coefficient of Variation

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Range (全距)

- The range of a data set is the difference between the largest and smallest data value.
- It is the simplest measure of variability.
- It is very sensitive to the smallest and largest data values.

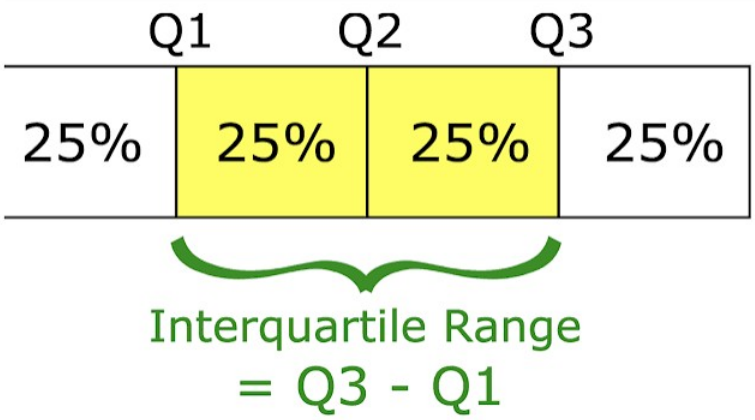
  Range = largest value – smallest value = 715 – 525 = 190.

525	530	530	535	535	535	535	535	540	540
540	540	540	545	545	545	545	545	550	550
550	550	550	550	550	560	560	560	565	565
565	570	570	572	575	575	575	580	580	580
580	585	590	590	590	600	600	600	600	610
610	615	625	625	625	635	649	650	670	670
675	675	680	690	700	700	700	700	715	715

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Interquartile Range (IQR)(四分位距)

- The interquartile range of a data set is the difference between the third quartile and the first quartile.
- It is the range for the middle 50% of the data.
- It overcomes the sensitivity to extreme data values.



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Interquartile Range (IQR)

- The interquartile range of a data set is the difference between the third quartile and the first quartile.
- It is the range for the middle 50% of the data.
- It overcomes the sensitivity to extreme data values.

3rd Quartile (Q_3) = 625

1st Quartile (Q_1) = 545

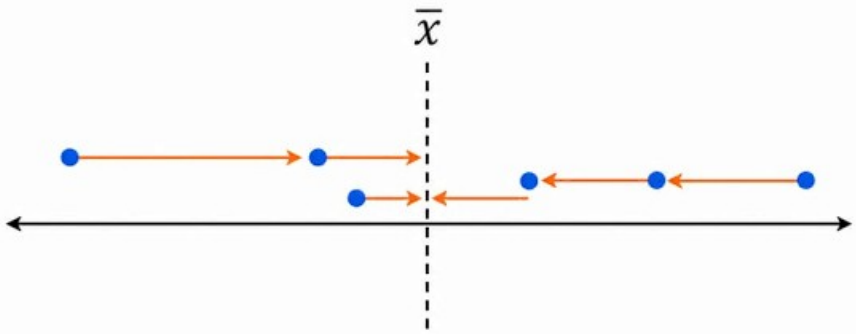
IQR = 625 – 545 = 80

525	530	530	535	535	535	535	535	540	540
540	540	540	545	545	545	545	545	550	550
550	550	550	550	550	560	560	560	565	565
565	570	570	572	575	575	575	580	580	580
580	585	590	590	590	600	600	600	600	610
610	615	625	625	625	635	649	650	670	670
675	675	680	690	700	700	700	700	715	715

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Variance (變異數)

- The variance is a measure of variability that utilizes all the data.
- It is based on the difference between the value of each observation (x_i) and the mean ($\bar{x}$ for a sample, μ for a population)➔ A deviation from the mean (離差)
- The variance is useful in comparing the variability of two or more variables.
- The variance is the average of the squared deviations between each data value and the mean.

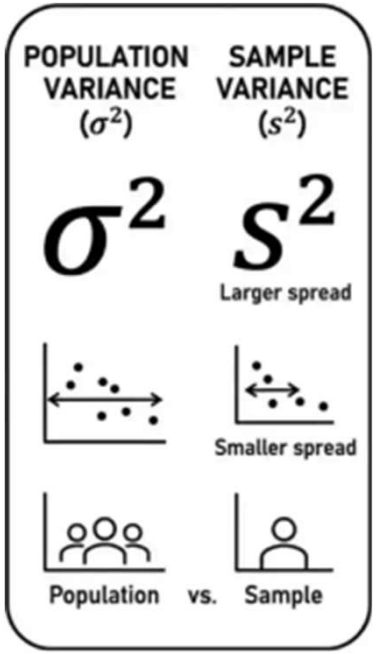


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Variance

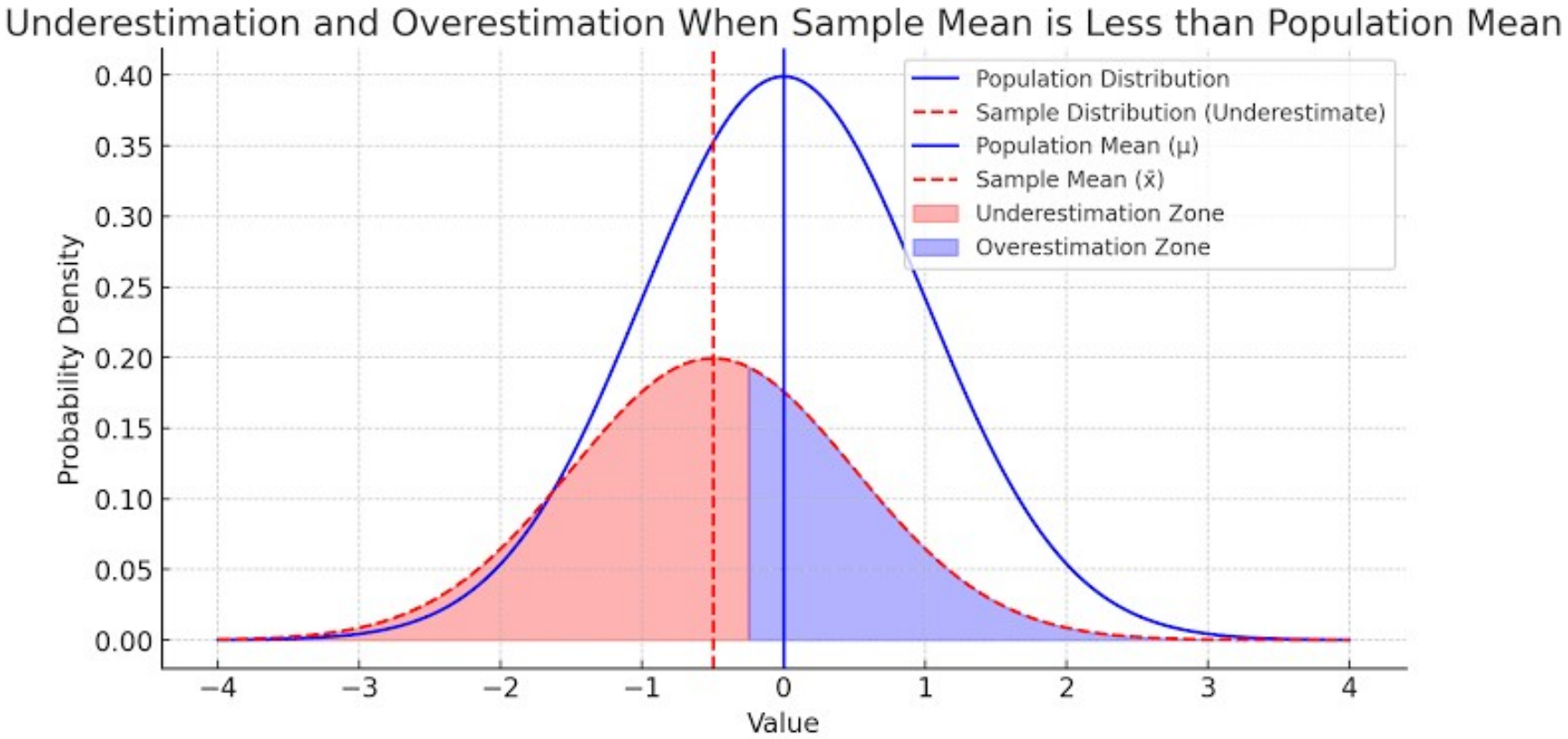
- The variance of a sample is: $$s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$$

- The variance for a population is: $$\sigma^2 = \frac{\sum(x_i - \mu)^2}{N}$$



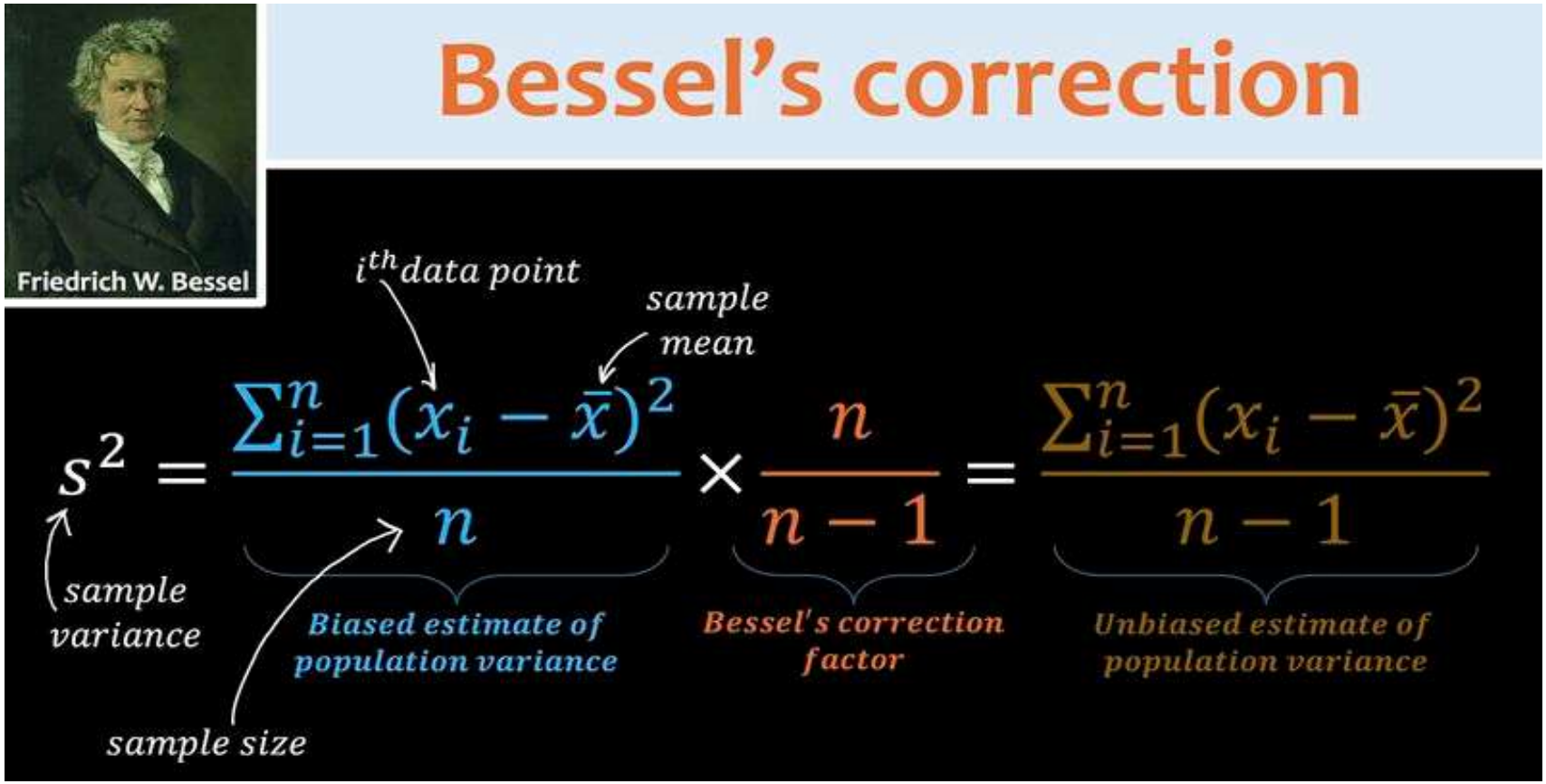
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Bessel's Correction (貝塞爾修正)



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Standard Deviation (標準差)

- The standard deviation of a data set is the positive square root of the variance.
- It is measured in the same units as the data, making it more easily interpreted than the variance.
- The standard deviation of a sample is: $s = \sqrt{s^2}$
- The standard deviation of a population is: $\sigma = \sqrt{\sigma^2}$

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Coefficient of Variation (變異係數)

- The coefficient of variation indicates how large the standard deviation is in relation to the mean.

➔ A measure of relative variability (相對變異程度)

Data Set A

- Mean=$100
- Standard deviation =$10



Data Set B

- Mean=$1000
- Standard deviation =$50

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Coefficient of Variation (變異係數)

- The coefficient of variation indicates how large the standard deviation is in relation to the mean.

➔ A measure of relative variability (相對變異程度)



Data Set A

- Mean=$100
- Standard deviation =$10

10% of its mean

Data Set B

- Mean=$1000
- Standard deviation =$50

5% of its mean

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Coefficient of Variation (變異係數)

- The coefficient of variation of a sample is: $\left[\frac{s}{\bar{x}} \text{ x } 100\right]\%$
- The coefficient of variation of a population is: $\left[\frac{\sigma}{\mu} \text{ x } 100\right]\%$

Lower CV ➔ Less variability relative to the mean
Higher CV ➔ More variability relative to the mean

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Sample Variance, Standard Deviation, and Coefficient of Variation

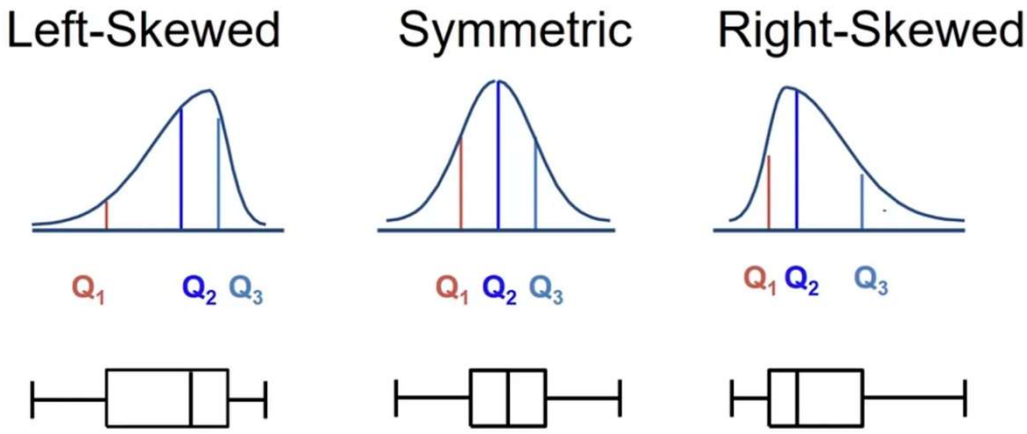
Example: Apartment Rents

- The variance is: $s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1} = 2{,}996.16$
- The standard deviation is: $s = \sqrt{s^2} = \sqrt{2{,}996.16} = 54.74$
- The coefficient of variation is: $\left[\frac{s}{\bar{x}} \text{ x } 100\right]\% = \left[\frac{54.74}{590.80} \text{ x } 100\right]\% = 9.27\%$

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Measures of Distribution Shape, Relative Location, and Detecting Outliers

- Distribution Shape
- z-Scores
- Chebyshev's Theorem
- Empirical Rule
- Detecting Outliers



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Distribution Shape: Skewness (1 of 2)

- An important numerical measure of the shape of a distribution is called skewness.
- The formula for the skewness of sample data is

$$\text{Skewness} = \frac{n}{(n-1)(n-2)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3$$

- Skewness can be easily computed using statistical software.

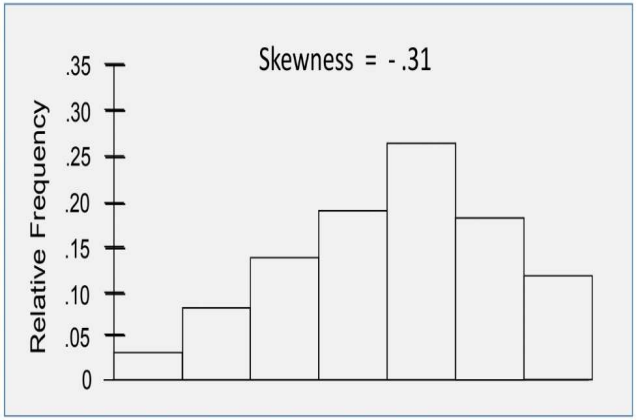
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Distribution Shape: Skewness (2 of 2)

Moderately Skewed Left

Skewness is negative.

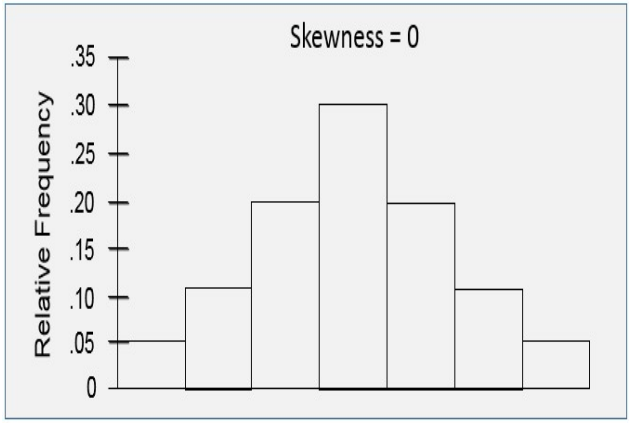
Mean will usually be less than the median.



Symmetric (not skewed)

Skewness is zero.

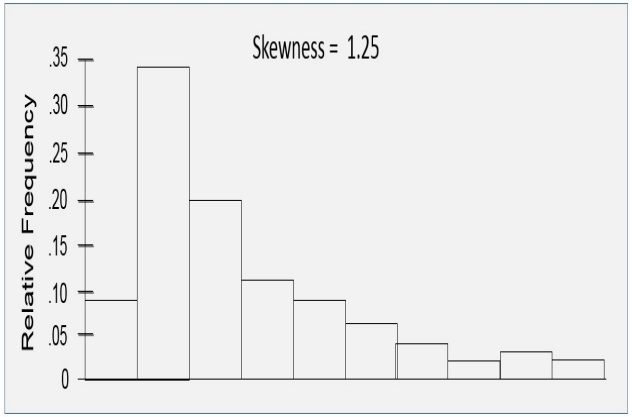
Mean and median are equal.



Highly Skewed Right

Skewness is positive (often above 1.0)

Mean will usually be more than the median.



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z-Scores (1 of 2)

- The z-score is often called the standardized value.
- It denotes the number of standard deviations a data value x_i is from the mean.
- An observation's *z*-score is a measure of the relative location of the observation in a data set.
- A data value less than the sample mean will have a *z*-score less than zero.
- A data value greater than the sample mean will have a *z*-score greater than zero.
- A data value equal to the sample mean will have a *z*-score of zero.

$$z_i = \frac{x_i - \bar{x}}{s}$$

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z-Scores (2 of 2)

Example: Apartment Rents

z-Score of Smallest Value (525)

$$z_i = \frac{x_i - \bar{x}}{s} = \frac{525 - 590.80}{54.74} = -1.20$$

Standardized Values for Apartment Rents

-1.20	-1.11	-1.11	-1.02	-1.02	-1.02	-1.02	-1.02	-0.93	-0.93
-0.93	-0.93	-0.93	-0.84	-0.84	-0.84	-0.84	-0.84	-0.75	-0.75
-0.75	-0.75	-0.75	-0.75	-0.75	-0.56	-0.56	-0.56	-0.47	-0.47
-0.47	-0.38	-0.38	-0.34	-0.29	-0.29	-0.29	-0.20	-0.20	-0.20
-0.20	-0.11	-0.01	-0.01	-0.01	0.17	0.17	0.17	0.17	0.35
0.35	0.44	0.62	0.62	0.62	0.81	1.06	1.08	1.45	1.45
1.54	1.54	1.63	1.81	1.99	1.99	1.99	1.99	2.27	2.27

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Chebyshev's Theorem (1 of 2)

- At least $(1 - 1/z^2)$ of the data values must be within z standard deviations of the mean, where z is any value greater than 1.
- Chebyshev's theorem requires $z > 1$; but z need not be an integer.
- At least 75% of the data values must be within $z = 2$ standard deviations of the mean.
- At least 89% of the data values must be within $z = 3$ standard deviations of the mean.
- At least 94% of the data values must be within $z = 4$ standard deviations of the mean.

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Chebyshev's Theorem (2 of 2)

Example: Apartment Rents

Let $z = 1.5$ with $\bar{x} = 590.80$ and $s = 54.74$

At least $(1 - 1/(1.5)^2) = 1 - 0.44 = 0.56$ or 56%

of the rent values must be between

$\bar{x} - z(s) = 590.80 - 1.5(54.74) =$ 509

and

$\bar{x} + z(s) = 590.80 + 1.5(54.74) =$ 673

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Empirical Rule (1 of 2)

When the data are believed to approximate a bell-shaped distribution:

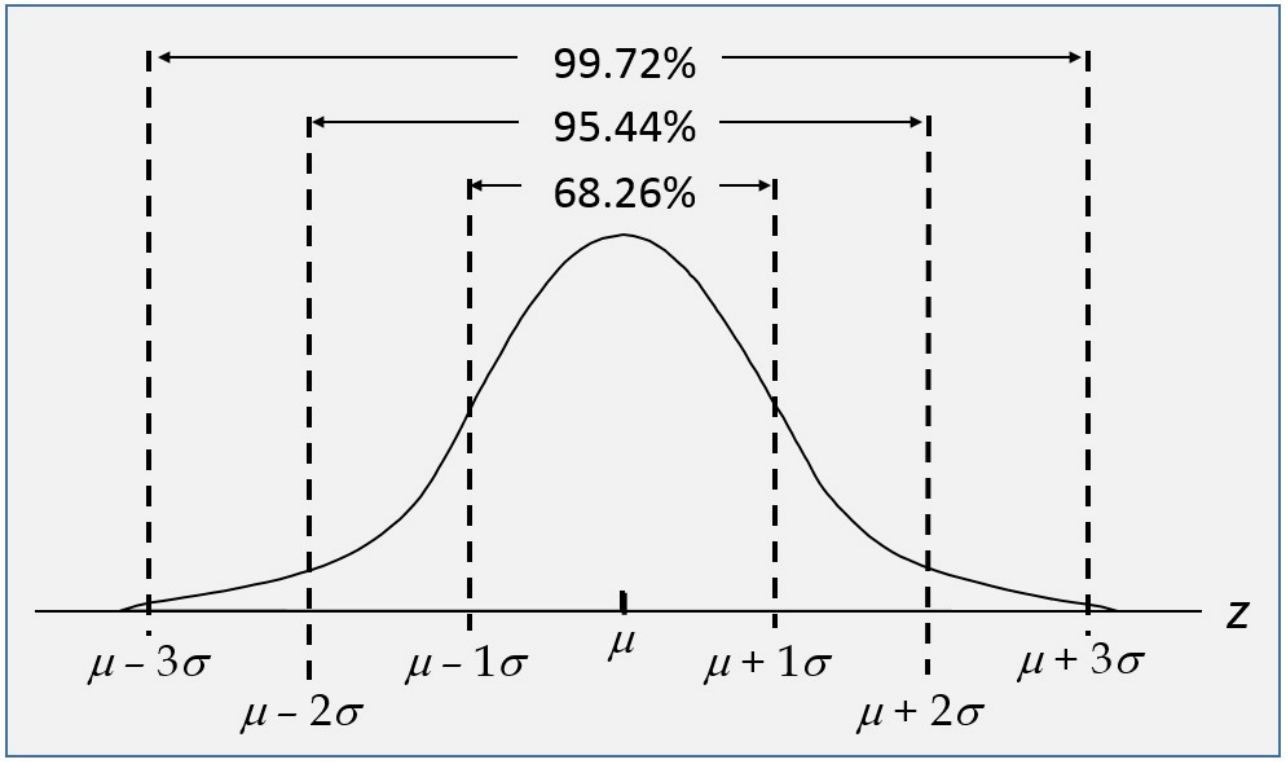
- The empirical rule can be used to determine the percentage of data values that must be within a specified number of standard deviations of the mean.
- The empirical rule is based on the normal distribution, which is covered in Chapter 6.

For data having a bell-shaped distribution:

- Approximately 68% of the data values will be within one standard deviation of the mean.
- Approximately 95% of the data values will be within two standard deviations of the mean.
- Almost all of the data values will be within three standard deviations of the mean.

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Empirical Rule (2 of 2)



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Detecting Outliers (1 of 2)

- An outlier is an unusually small or unusually large value in a data set.
- A data value with a z-score less than –3 or greater than +3 might be considered an outlier.
- It might be:
  - an incorrectly recorded data value
  - a data value that was incorrectly included in the data set
  - a correctly recorded data value that belongs in the data set

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Detecting Outliers (2 of 2)

Example: Apartment Rents

- The most extreme z-scores are -1.20 and 2.27.
- Using $|z| \geq 3$ as the criterion for an outlier, there are no outliers in this data set.

Standardized Values for Apartment

-1.20	-1.11	-1.11	-1.02	-1.02	-1.02	-1.02	-1.02	-0.93	-0.93
-0.93	-0.93	-0.93	-0.84	-0.84	-0.84	-0.84	-0.84	-0.75	-0.75
-0.75	-0.75	-0.75	-0.75	-0.75	-0.56	-0.56	-0.56	-0.47	-0.47
-0.47	-0.38	-0.38	-0.34	-0.29	-0.29	-0.29	-0.20	-0.20	-0.20
-0.20	-0.11	-0.01	-0.01	-0.01	0.17	0.17	0.17	0.17	0.35
0.35	0.44	0.62	0.62	0.62	0.81	1.06	1.08	1.45	1.45
1.54	1.54	1.63	1.81	1.99	1.99	1.99	1.99	2.27	2.27

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Five-Number Summaries and Box Plots

- Summary statistics and easy-to-draw graphs can be used to quickly summarize large quantities of data.
- Two tools that accomplish this are five-number summaries and box plots.

1. Smallest Value
2. First Quartile
3. Median
4. Third Quartile
5. Largest Value

$$L_p = \left(\frac{p}{100}\right)(n+1)$$

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Five-Number Summary

Example: Apartment Rents

Lowest Value = 525 First Quartile = 545

Median = 575

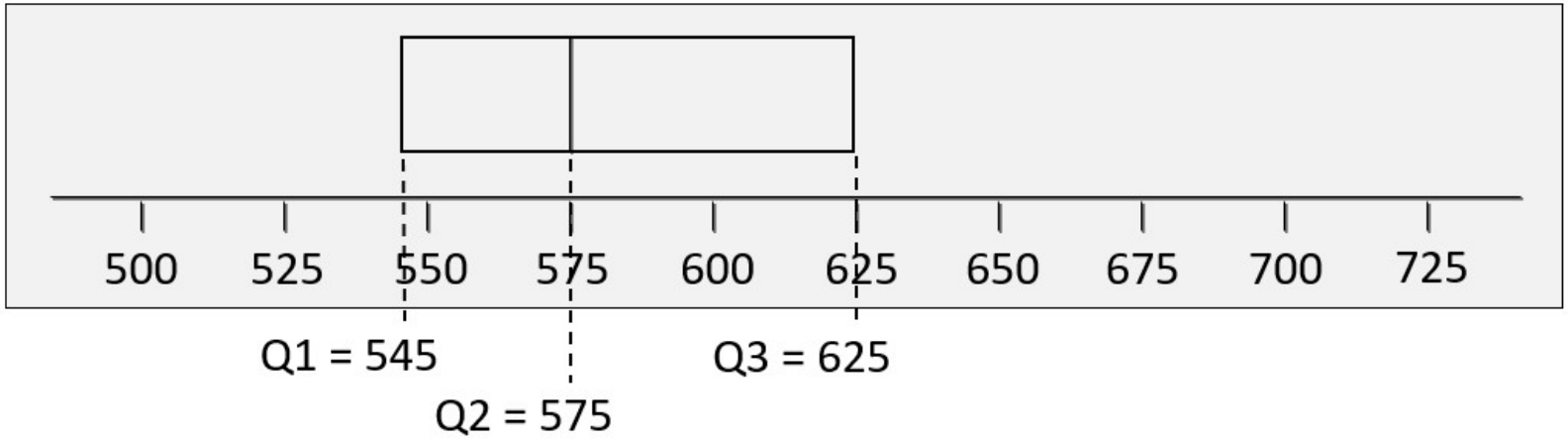
Third Quartile = 625 Largest Value = 715

525	530	530	535	535	535	535	535	540	540
540	540	540	545	545	545	545	545	550	550
550	550	550	550	550	560	560	560	565	565
565	570	570	572	575	575	575	580	580	580
580	585	590	590	590	600	600	600	600	610
610	615	625	625	625	635	649	650	670	670
675	675	680	690	700	700	700	700	715	715

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Box Plot

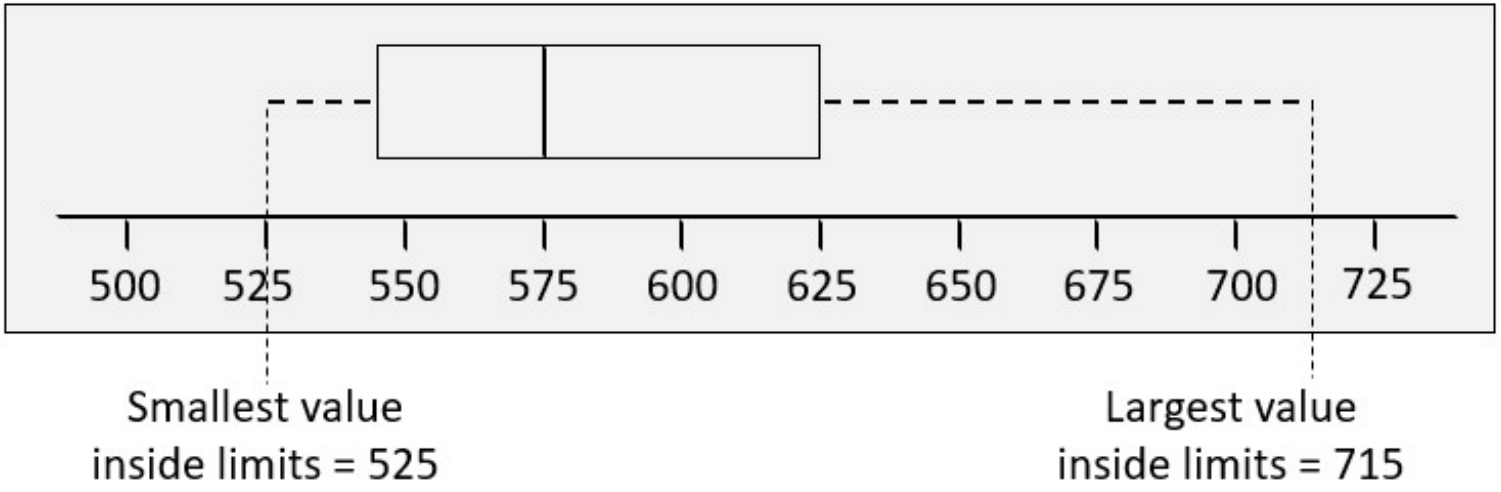
- A box plot is a graphical display of data that is based on a five-number summary.
- Box plots provide another way to identify outliers.
- A box is drawn with its ends located at the first and third quartiles.
- A vertical line is drawn in the box at the location of the median (second quartile).



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Box Plot

- Limits are located (not drawn) using the interquartile range (IQR).
- Data outside these limits are considered outliers.
- The location of each outlier is shown with the symbol * .



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Box Plot

Example: Apartment Rents

- The lower limit is located 1.5(IQR) below Q_1.

  Lower Limit: $Q_1 - 1.5(\text{IQR}) = 545 - 1.5(80) = 425$

- The upper limit is located 1.5(IQR) above Q_3.

  Upper Limit: $Q_3 + 1.5(\text{IQR}) = 625 + 1.5(80) = 745$

- There are no outliers (values less than 425 or greater than 745) in the apartment rent data.

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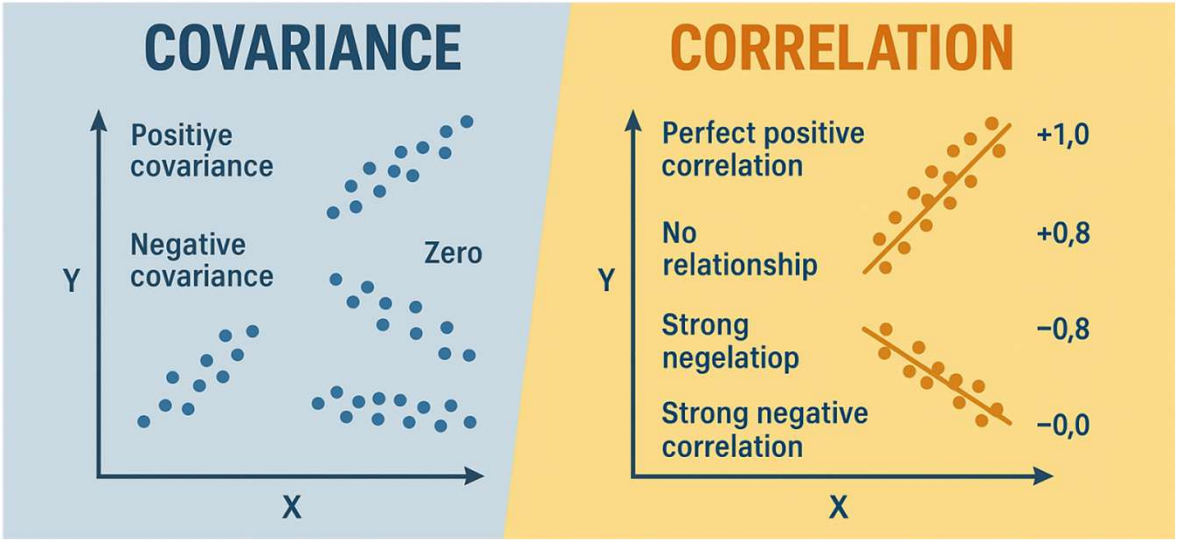
Measures of Association Between Two Variables

- Thus far we have examined numerical methods used to summarize the data for one variable at a time.
- Often a manager or decision maker is interested in the relationship between two variables.
- Two descriptive measures of the relationship between two variables are covariance and correlation coefficient.

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Covariance (共變異數)

- The covariance is a measure of the linear association between two variables.
- Positive values indicate a positive relationship.
- Negative values indicate a negative relationship.



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Covariance (共變異數)

- The covariance is a measure of the linear association between two variables.
- Positive values indicate a positive relationship.
- Negative values indicate a negative relationship.
- The covariance is computed as follows:

For samples: $$s_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

For populations: $$\sigma_{xy} = \frac{\sum(x_i - \mu_x)(y_i - \mu_y)}{N}$$

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Correlation Coefficient (相關係數)

- Correlation is a measure of linear association and not necessarily causation.
- Just because two variables are highly correlated, it does not mean that one variable is the cause of the other.
- The correlation coefficient is computed as follows:

For samples: $r_{xy}=\frac{s_{xy}}{s_x s_y}$

For populations: $\rho_{xy}=\frac{\sigma_{xy}}{\sigma_x \sigma_y}$

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Correlation Coefficient

- The coefficient can take on values between –1 and +1.
- Values near –1 indicate a strong negative linear relationship.
- Values near +1 indicate a strong positive linear relationship.
- The closer the correlation is to zero, the weaker the relationship.

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Covariance and Correlation Coefficient (1 of 3)

A golfer is interested in investigating the relationship, if any, between driving distance and 18-hole score.

Average Driving Distance (yards)	Average 18-Hole Score
277.6	69
259.5	71
269.1	70
267.0	70
255.6	71
272.9	69

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Covariance and Correlation Coefficient (2 of 3)

Example: Golfing Study

	x	y	$(x_i-\bar{x})$	$(y_i-\bar{y})$	$(x_i-\bar{x})(y_i-\bar{y})$
	277.6	69	10.65	-1.0	-10.65
	259.5	71	-7.45	1.0	-7.45
	269.1	70	2.15	0	0
	267.0	70	0.05	0	0
	255.6	71	-11.35	1.0	-11.35
	272.9	69	5.95	-1.0	-5.95
Average	267.0	70.0		Total	-35.40
Std. Dev.	8.2192	.8944			

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Covariance and Correlation Coefficient (3 of 3)

Example: Golfing Study

- Sample Covariance: $s_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n-1} = \frac{-35.40}{6-1} =$ -7.08

- Sample Correlation Coefficient: $r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{-7.08}{(8.2192).8944)} =$ −.9631

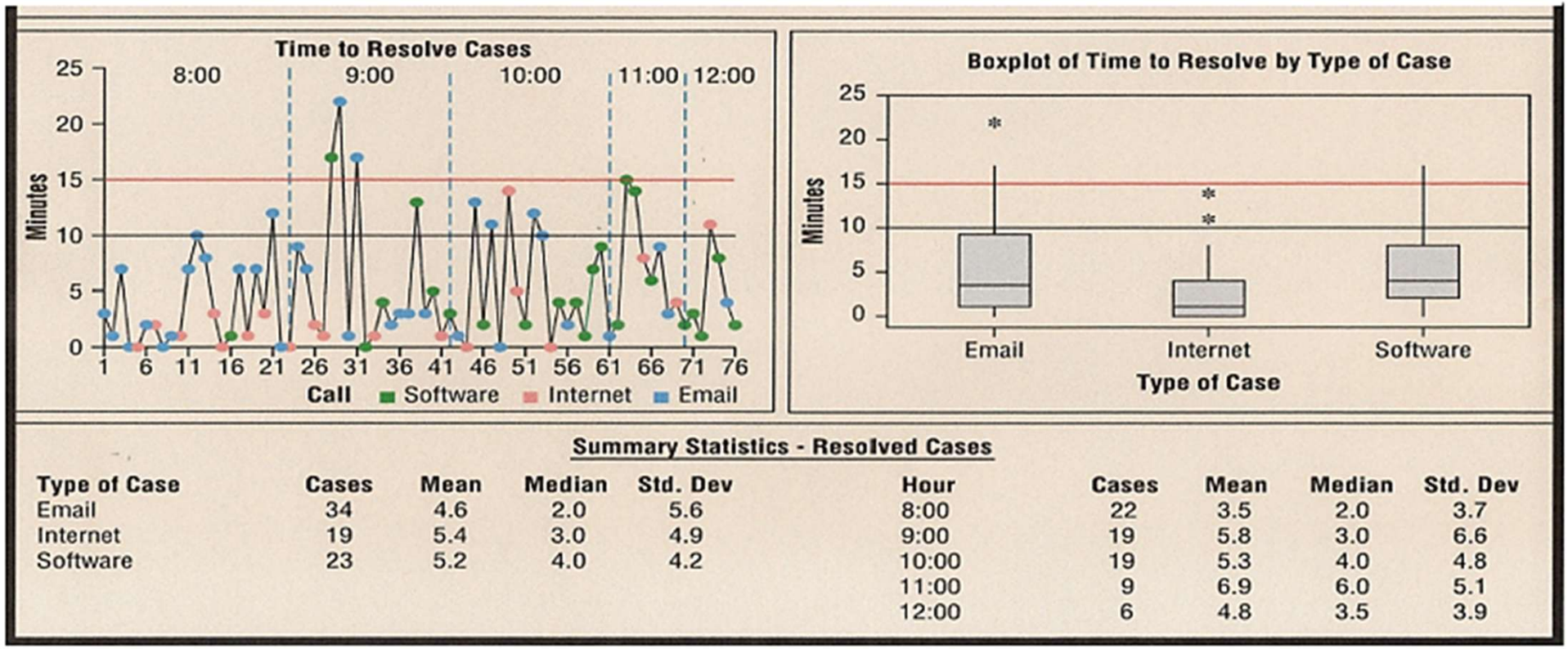
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Data Dashboards: Adding Numerical Measures to Improve Effectiveness

- Data dashboards are not limited to graphical displays.
- The addition of numerical measures, such as the mean and standard deviation of KPIs, to a data dashboard is often critical.
- Dashboards are often interactive.
- Drilling down refers to functionality in interactive dashboards that allows the user to access information and analyses at an increasingly detailed level.

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Data Dashboards: Adding Numerical Measures to Improve Effectiveness (2 of 2)



Summary Statistics - Resolved Cases

Type of Case	Cases	Mean	Median	Std. Dev
Email	34	4.6	2.0	5.6
Internet	19	5.4	3.0	4.9
Software	23	5.2	4.0	4.2

Hour	Cases	Mean	Median	Std. Dev
8:00	22	3.5	2.0	3.7
9:00	19	5.8	3.0	6.6
10:00	19	5.3	4.0	4.8
11:00	9	6.9	6.0	5.1
12:00	6	4.8	3.5	3.9

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